

Barret Schloerke

```
code[c("architect", "manager", "philosopher")]
data[c("wrangler", "explorer", "visualizer")]
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Education

2013
2017

Doctor of Philosophy in Statistics, *Purdue University*, West Lafayette, IN

Advisors: Dr. William Cleveland and Dr. Ryan Hafen; Emphasis: Large Data Visualization

National Science Foundation Graduate Research Fellowship Recipient (2012-2017)

2013
2014

Master of Science in Mathematical Statistics, *Purdue University*, West Lafayette, IN, 3.69/4.0

2012
2012

Doctor of Philosophy in Statistics, *Rice University*, Houston, TX

Advisor: Dr. Hadley Wickham

2006
2010

Bachelor of Science in Computer Engineering, *Iowa State University*, Ames, IA, 3.77/4.0

Work Experience

2018
2021

RStudio, Software Engineer

- Core developer of *Shiny*, an R package that makes it easy to build interactive web apps straight from R.
- Architected, developed, and automated the testing of 200+ *Shiny* applications using GitHub Actions over multiple operating systems and R versions.

2011
2012

Metamarkets, *San Francisco, CA*, Software Engineer

Client-Side (CoffeeScript)

Front-end Server (Node.js & CoffeeScript)

- | | |
|--|---|
| <ul style="list-style-type: none">○ Implemented URL routing framework which displayed consistent URLs between users○ Implemented DVL visualizations using D3 for all interactive data visualizations○ Utilized DVL library to make all variables reactive to dependency value changes○ Modularized code to be reusable/plugin-in-play○ Reduced website startup time for faster loading○ Trimmed data queries to grab new data only to reduce unnecessary server load○ Refined anomaly detection algorithms to highlight visual anomalies | <ul style="list-style-type: none">○ Automated server deployment process to allow any engineer to deploy services○ Created server cluster to prevent request failure○ Developed company/user management system○ Developed configuration files to determine where and how data is displayed○ Integrated configuration files to help automate self-serve data processing○ Maintained client security○ Communicated with multiple back-end services to maintain a consistent client interface |
|--|---|

2016
2018

Gates Foundation: HBGDKi, *Remote*, Tool Development Team

- Provided customized summary statistics in timely manner for 75+ datasets to feed into visualization applications
- Maintained data security while coordinating with five other data scientists
- Created and maintained shell R packages for standardized R package development

2016
2017

DARPA: XDATA and D3M Grants, *Purdue University*

- Collaborated remotely with Hafen Consulting, Kitware, and KnowledgeVis to produce a web interface for SMEs to interact with machine learning pipelines created by other working groups
- Completed multiple miniature hack-a-thons with my team to present consistent findings of our results
- Served as lead point-of-contact for integrating services with other working groups

2017
2017

Big Data Analytics: Statistics and Data Visualization, *Purdue University*, Teaching Assistant

- Taught several lectures to help students understand how to use R for *big* data
- Explored easy to understand concepts in class that have difficulty scaling to larger data

2012
2015

Hadley Wickham R Master Class, *San Francisco, CA and Chicago, IL*, Teaching Assistant

- Fielded advanced programming technique questions from students to solidify content presented
- Aided students in developing R packages on multiple computing platforms

2008
2010

Novartis Pharmaceutical R&D, Basel, Switzerland, Data Scientist Intern

- Created a time management tool using gantt charts to help management quickly display employee availability
- Created a web-based tutorial consisting of examples of the most common graphics produced when using different layouts by “white-box” ing complex visualizations into reusable plotting layers
- Worked cohesively with the top 15 statisticians within Novartis

Honors and Awards

2012
2017

National Science Foundation Graduate Research Fellowship

- Received a three-year fellowship (to be used within five years) to pursue graduate studies in a science based field
- Only awardee for Computational Statistics of the thirteen statistical recipients in 2012
- One of five recipients from Iowa State University in 2012

2015
2015

Visiting Scientific Researcher, *Monash University*, Melbourne, Victoria, Australia

- Presented latest research on `gqlr` to WOMBAT2016
- Collaborated with NUMBAT on large data exploration tools

2011/6

Best Dynamic / Interactive Visualization, *Data Insight (SF)*, San Francisco, CA

- Team of four worked together over the weekend to produce an interactive visualization
- Leaned on each other's skills to provide an application that highlighted our data and visualization talents

2011/4

National Science Foundation Graduate Research Fellowship, *Honorable mention*

Publications

Journal Articles

2017

Gökalp, F., Barret Schloerke. "Parallel Programming in Linear Mixed Models." *The R Journal*, Submitted 08/2017.

2016

Schloerke, B., Hadley Wickham, Dianne Cook, Heike Hofmann. "Escape from Boxland: Generating a Library of High-Dimensional Geometric Shapes." *The R Journal*, 8(2):243-257, December 2016.

2013

Emerson, J., W. Green, B. Schloerke, J. Crowley, D. Cook, H. Hofmann, and H. Wickham "The Generalized Pairs Plot." *Journal of Computational and Graphical Statistics* 22.1 (2013). Print.

2012

Hofmann H., D. Cook, C. Kielion, B. Schloerke, J. Hobbs, A. Loy, L. Mosley, D. Rockoff, Y. Huang, D. Wrolstad, and T. Yin "Delayed, Canceled, on Time, Boarding... Flying in the USA." *Journal of Computational and Graphical Statistics* 20.2 (2012). Print.

2010

Mosley, L., D. Cook, H. Hofmann, C. Kielion, and B. Schloerke. "Monitoring the 2008 Election Visually." *Chance* 23.3 (2010). Print.

Presentations

2024/8

Schloerke, B. "Editable data frames in Shiny for Python" posit::conf. Posit. Aug. 2024. Speech.

2024/6

Schloerke, B. "shinylive: Serverless Shiny apps" YouTube. Posit. Jun. 2024. Video.

2024/3

Schloerke, B. "shinylive: Serverless Shiny apps" Shiny Conference. Appsilon. May. 2024. Speech.

2023/3

Schloerke, B. "Lessons learned from testing 2500+ Shiny Apps every day" Shiny Conference. Appsilon. May. 2023. Speech.

2022/10

Schloerke, B. "shinytest2: Regression Testing for Shiny Applications" Shiny Day. Johnson and Johnson. Oct. 2022. Speech.

2022/7

Schloerke, B. "shinytest2: Regression Testing for Shiny Applications" rstudio::conf. RStudio. Forest Heights, MD. July. 2022. Speech.

2022/6

Schloerke & Mostipak. "Programming Games with Shiny — Roll the Dice: with Quosures!" YouTube. Posit. Jun. 2022. Video.

2022/5

Schloerke & Mostipak. "Programming Games with Shiny — Roll the Dice" YouTube. Posit. May. 2022. Video.

2022/5

Schloerke & Mostipak. "Programming Games with Shiny — Guess the Number" YouTube. Posit. May. 2022. Video.

2022/5

Schloerke & Mostipak. "Programming Games with Shiny — Dragon Realm" YouTube. Posit. May. 2022. Video.

2022/5

Schloerke, B. "shinytest2: Regression Testing for Shiny Applications" R in Pharma working group. May. 2022. Webinar.

2022/4 Schloerke, B. "shinytest2: Regression Testing for Shiny Applications" Shiny Conference. Appsilon. Apr. 2022. Speech.

2022/2 Schloerke, B. "Maximize computing resources using future_promise" YouTube. Posit. Feb. 2022. Video.

2021/8 Schloerke, B. "Plumber: Asynchronous Route Execution with Barret Schloerke" YouTube. Harvard Data Science Initiative. Aug. 2021. Webinar.

2021/3 Blaire & Schloerke. "Plumber: Asynchronous Route Execution with Barret Schloerke" RStudio Webinar. Mar. 2021. Webinar.

2021/1 Schloerke, B. "plumber + future: Async Web APIs" rstudio::global. Jan. 2021. Speech.

2020/8 Schloerke, B. "Episode 14: Shining a Light on learnr (Barret Schloerke Part 3)" Shiny Developer Series with Eric Nantz. Aug. 2020. Podcast.

2020/8 Schloerke, B. "Episode 13: Inside Plumber 1.0 (Barret Schloerke Part 2)" Shiny Developer Series with Eric Nantz. Aug. 2020. Podcast.

2020/8 Schloerke, B. "Episode 12: Barret Schloerke Part 1 (reactlog)" Shiny Developer Series with Eric Nantz. Aug. 2020. Podcast.

2020/04 Schloerke, B. "Reactlog 2.0: Debugging the State of Shiny" Remote Meetup: Shiny Deep Dive. Statistical Programming DC. Washington D.C. Apr. 2020. Speech.

2019/11 Schloerke, B. "shinyrmeta: Reproducible Shiny apps with shinyrmeta" Autumn Biostatistics Conference US (ABACUS). GlaxoSmithKline. Phoenixville, PA. Nov. 2019. Speech.

2019/5 Schloerke, B. "Debug Large Data Apps: shiny + reactlog + sparklyr" Advanced R workshop. Northeastern University. Boston, MA. Mar. 2019. Speech.

2019/1 Schloerke, B. "Reactlog 2.0: Debugging the State of Shiny" rstudio::conf. RStudio. Austin, TX. Jan. 2019. Speech.

2018/8 Schloerke, B. "Shiny" DATANAUTS. NASA. Washington, D.C. Aug. 2018. Speech.

2017/4 Schloerke, B. "Cognostics: Metrics enabling detailed interactive visualization of big data" CSESC2017. Purdue University, West Lafayette, IN. Apr. 2017. Speech.

2017/1 Schloerke, B. "tidyverse[c(\"magrittr\", \"dplyr\", \"tidyr\")]" Purdue Graduate Statistics Seminar. Math, West Lafayette, IN. Jan. 2017. Speech.

2016/8 Schloerke, B. "Cognostics: Metrics enabling detailed interactive visualization of big data" Joint Statistical Meeting. Chicago, IL. Aug. 2016. Speech.

2016/6 Schloerke, B. "ggduo: Pairs plot for two group data" UseR!2016. Stanford University, Stanford, CA. Jun. 2016. Speech.

2016/6 Schloerke, B. "High Performance Computing for High-Resolution Analysis of Big Data and Small" Spring Research Conference. Illinois Institute of Technology, Chicago, IL. Jun. 2016. Speech.

2016/4 Schloerke, B. "Analysis and Visualization of Large Complex Data with Tessera". Graduate Statistics Seminar. Purdue, West Lafayette, IN. Mar. 2016. Speech.

2016/3 Schloerke, B. "Analysis and Visualization of Large Complex Data with Tessera". Graphics Research Group Seminar. Iowa State University, Ames, IA. Mar. 2016. Speech.

2016/2 Schloerke, B. "Analysis and Visualization of Large Complex Data with Tessera". NUMBAT Seminar. Monash University, Melbourne, VIC AUS. Feb. 2016. Speech.

2016/2 Schloerke, B. "GraphQL: An R server implementation". Wombat2016. Melbourne Zoo, Melbourne, VIC AUS. Feb. 2016. Speech.

2015/10 Schloerke, B. "Web Scraping with R" American Credit Acceptance. Spartanburg, SC. Oct. 2015. Speech.

2015/2 Schloerke, B. "Trelliscope: D&R visualization tool" Big Data Visualization. Fields Institute Toronto, CAN. Feb. 2015. Speech.

2014/10	Schloerke, B. "ggplot2: displaying spatial and temporal data" Spatial Statistics Seminar. Recitation, West Lafayette, IN. Oct. 2014. Speech.
2014/2	Schloerke, B. "Reducing Working Environment Inefficiencies" Graduate Statistics Seminar. Haas, West Lafayette, IN. Feb. 2014. Speech.
2010/9	Schloerke, B. "helpR: Help for R." Working Group Meeting. Snedecor Hall, Ames, IA. Sept. 2010. Speech.
2010/8	Schloerke, B. "GGally: A Plot Matrix for All Variable Types." Joint Statistical Meeting 2010. Vancouver Convention Center, Vancouver, BC Canada. Aug. 2010. Speech.
Lightning Talks	
2014/6	"git", Purdue Working Group
2011	"Sinartra: Simple Web Framework", "5 Tips for Running Node in Production", "Express: Node Router", San Francisco Bay Area

Projects

Posit

Advisors:	Joe Cheng and Winston Chang
2023 2024	shiny.render.data_frame, Editable data frames within Shiny for Python applications, <i>Posit</i>
2023 2024	shiny.render.renderer, Extendable base class that creates renderers within Shiny for Python applications, <i>Posit</i>
2023	shiny.playwright, Playwright testing controllers for every Shiny for Python UI element, <i>Posit</i>
2023	shiny.pytest, Pytest integration for testing Shiny for Python applications, <i>Posit</i>
2023	shinylive for R , Export Shiny for R applications to a static folder to be hosted statically, <i>Posit</i>
2023 2023	shinylive for Python , Export Shiny for Python applications to a static folder to be hosted statically, <i>Posit</i>
2023 2024	shinylive assets , Assets required to run Shiny applications in the browser, <i>Posit</i>
2023	htmltools, Bootswatch + Bootstrap 5 themes for Shiny for Python, <i>Posit</i>
2023	shinyswatch, Bootswatch + Bootstrap 5 themes for Shiny for Python, <i>Posit</i>
2023	shiny, A web development framework for Python, <i>Posit</i>

RStudio

Advisors:	Joe Cheng and Winston Chang
2019	shinycoreci, Regression testing for shiny-verse, <i>RStudio</i>
2018	shiny, Easy interactive web applications with R, <i>RStudio</i>

2021 2023	htmltools, Tools for HTML, <i>RStudio</i>
2021 2023	shinytest2, Testing for Shiny Applications, <i>RStudio</i>
2021 2022	chromote, Headless Chrome Web Browser Interface, <i>RStudio</i>
2018 2022	swagger, Dynamically Generates Documentation from a 'Swagger' Compliant API, <i>RStudio</i>
2021 2021	plumberDeploy, Take screenshots of webpages from R, <i>RStudio</i>
2020 2021	plumberDeploy, Plumber Deployment, <i>RStudio</i>
2018 2019	gt, Easily Create Presentation-Ready Display Tables, <i>RStudio</i>
2018 2022	plumber, Turn your R code into a web API., <i>RStudio</i>
2018 2020	reactlog, Reactivity Visualizer for 'shiny', <i>RStudio</i>
2018 2018	leaflet, R Interface to Leaflet Maps, <i>RStudio</i>

Purdue University

Advisors: Dr. Ryan Hafen and Dr. William Cleveland

2017 2021	autocogs, Automatic Cognostic Calculations, <i>Purdue University</i> ○ "Automatically calculates cognostics for plot objects and list column plot objects. autocogs compliments trelliscopejs's panel interactions by producing multiple cognositc values for the visualizations displayed" ○ Generalized framework to produce consistent cognostics independent of visualization library utilized
2017 2015	gqlr, GraphQL Server in R, <i>Purdue University</i> ○ "R server implementation of GraphQL, a query language created by Facebook for describing complex data queries independent of the storage format." ○ GraphQL provides a complete and human readable description of the data in your data schema and gives clients the power to query only for exactly what they need. ○ gqlr is a native R GraphQL implementation to be used with Relay in React javascript web applications
2016 2016	GGally::ggduo, Generalized plot matrix for two-grouped data, <i>Purdue University</i> ○ Extension of ggpairs built on top of a ggmatrix object ○ Pairs plot for two group data ○ Readily useful for multiple time series analysis and canonical correlation analysis ○ Foundation function for 'ggnostic', a model diagnostic plot matrix, and 'ggts', a time series plot matrix ○ Funded by Google Summer of Code with final report
2015 2016	GGally::ggmatrix, Generalized plot matrix, <i>Purdue University</i> ○ Handles arbitrary plot columns and rows ○ Used as display mechanism for all GGally plot matrix functions
2015 2017	trelliscopejs, Create Interactive Trelliscope Displays, <i>Purdue University</i> ○ "Trelliscope is a scalable, flexible, interactive approach to visualizing data. This package provides methods that make it easy to create a Trelliscope display specification for trelliscopejs. High-level functions are provided for creating displays from within dplyr or ggplot2 workflows. Low-level functions are also provided for creating new interfaces." ○ Ported Shiny-based trelliscope R package to be built with React framework to increase interaction speed ○ Integrated with ggplot2 objects to seamlessly produce trelliscopejs applications

2016	2016	rbokeh , R interface to Bokeh, <i>Purdue University</i>
		“A native R plotting library that provides a flexible declarative interface for creating interactive web-based graphics, backed by the Bokeh visualization library.” - Utilized non-standard evaluation within package to remove massive amounts of boilerplate
2013	2013	Bitcoin , <i>DARPA XDATA</i>
		- Brought previous transaction exporter up to protocol - Formatted escrow transactions to be matched easier - Made “None” keys uniquely identifiable, rather than being treated the same - Matched some “None” keys to valid output keys - Updated User exporter to include all information available
		Metamarkets
2012	2011	DVL , Data Visualization Legos, <i>Metamarkets</i>
		Functionally reactive JavaScript library to bind data to functions and visualizations
2012	2012	yalog , Yet Another Logger, <i>Metamarkets</i>
		Request logger to allow for personalized logging messages within different log levels
		Rice University
		Advisor: Dr. Hadley Wickham
2010	2010	helpR , <i>Help for R</i>
		Bettters Friendly HTML documentation with links to other packages, function aliases, and function sources. Finding information may be done with the comprehensive search bar.
2010	2010	VIGRE Program
		- Performed exploratory analysis on the housing crisis - Found, cleaned, and explored multiple data sets - Created all outputs reproducible from scratch
		Iowa State University Statistics Research
		Advisors: Dr. Dianne Cook and Dr. Heike Hofmann
	2010	GGally , <i>Extension to ggplot2, Iowa State University, Purdue University, and Google Summer of Code</i>
		“GGally extends ggplot2 by adding several functions to reduce the complexity of combining geometric objects with transformed data. Some of these functions include a pairwise plot matrix, a two group pairwise plot matrix, a parallel coordinates plot, a survival plot, and several functions to plot networks.” - Maintained R package with 25k+ monthly downloads - Collaborated with 10 major authors and many contributors - Assembled multiple plot matrix functions to aid in full data exploration - Integrated development process with <code>lintr</code> and <code>testthat</code> for code and output consistency
2010	2010	cranvas , <i>New Plotting Canvas for R</i>
		- New plotting canvas for R. Produces more than a million points in less than a second. - Dr. Wickham, Dr. Lawrence, Dr. Hoffman, Dr. Cook, Yihui Xie, Tengfei Yin, Marie Vendettuoli, Barret Schloerke
2009	2009	DescribeDisplay , <i>Reproduce GGobi images in R</i>
		Produce publication quality graphics from GGobi's describe display plugin.
2009	2009	tourr , <i>Geodesic interpolation and basis generation functions</i>
		- Implements geodesic interpolation and basis generation functions that allow you to create new tour methods in R. - Dr. Wickham, Dr. Cook, Barret Schloerke - Iowa State University and Rice University
2009	2009	American Statistical Association Data Expo , <i>Delayed, Cancelled, On-Time, Boarding ... Flying in the USA</i>
		- Displayed airport delays caused by airport layout using multiple applications. - Using Google's KML language, R and wget, we produced Google Earth images with the runways highlighted. - Iowa State University working group finished 2nd in the world competition.

2008
2008

Election Data, *Explore new ways to show the Electoral College for the 2008 Presidential Election*

- Created a custom cartogram of the United States with area proportional to the electoral vote of each state.
- Produced the 'Electoral Vote Towers' (inspired by New York Times 2000 Presidential Race) that quickly portrayed the electoral vote status.

2008
2008

DNA, *Viewing DNA sequence data*

- Extension of the polytope ranking.
- Created combinations of DNA sequences with a length of four which were displayed in a grand tour.

2008
2008

Polytope Ranking, *Visualizing ranked data*

- Created a model that would connect a ranked data set.
- Added jittered data points to each combination with the distance relative to the amount of points.

2007
2007

FaceOff, *'Lines and Dots' in the fourth dimension*

- Game created to calibrate users into viewing and thinking in the fourth dimension.
- Game is a combination of "Connect 4", "Tic Tac Toe", and "Dots and Lines."

2007
2007

IEEE Infovis Competition, *2007 Infovis visualization competition*

- Internet Movie Data Base (IMDB) competition.
- Worked on the genres of the database using the "Many Eyes" web application.
- Helped edit and produce the movie that the working group submitted.

2007
2007

GeoZoo, *A library of high dimension geometric objects*

- Functions to produce cubes, spheres, simplices, polyhedra, polytopes, tori, and mobius-like objects.
- Contains a new addition to the high dimension polytopes, a Hyper Ring Torus

Service

2010
2017

Clubs, *Iowa State University*, Developer

- Hip Hop Club (Dub H): Created an optimal sorting algorithm to place dancers into dances sans politics. Reduced semesterly data entry of 400+ records from 10 hours to 1 hour by allowing concurrent website inputs. Maintained current and historical semester attendance, waiver, and roster information for administrative purposes.
- Greek Week: Created an internet-based check-in system tied to university ID cards (2300+ members). Increased maximum check-in rate from one person every twenty seconds to one person every two seconds. Anonymity was maintained to prevent participation bullying within fraternities and sororities.

2017
2017

R for Data Science Seminar, *Purdue University*, Instructor

- Organized and presented course material in interactive weekly sessions
- Answered student's questions and adapted the presentations accommodate variable skill levels
- Coordinated with students maximize student attendance

2016
2017

Statistics Graduate Student Office, *Purdue University*, President

- Effectively ran a town hall meeting to positively address major concerns of statistics graduate students
- Organized "Graduate Student Mentor Program" at beginning of Fall Semester
- Organized "Esteemed Speaker" event with Dr. Arthur Dempster for the Spring semester

2014
2016

StatCom Assistant

- Coordinated and put on multiple events
- Helped in production of StatCom banner

2014
2016

Graduate Student Mentor

Technical

Languages

- Expert: R, Python, JavaScript, TypeScript, Node.js, GraphQL, Markdown, HTML, $\text{\LaTeX}$, JSON, YAML
- Moderate: MySQL, JSX, Regular Expressions, Bash, CSS, C

Systems

- Expert: Travis CI, GitHub, GitHub Pages, tidyverse.R, Hadoop
- Moderate: React.js, Drat.R, Broccoli.js, Webpack.js